Supplementary Notes: Conformal Prediction

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Note: this topic was covered in the sixteen-week version of this course. It is provided here as optional reading and is not part of the eight-week core. Nothing in these notes is required for the weekly assignments or the course project.

1 Introduction

Conformal prediction (CP) is a framework for **uncertainty quantification (UQ)** in machine learning. It wraps around any black-box model to produce prediction *sets* or *intervals* with rigorous, distribution-free coverage guarantees. Unlike Bayesian or ensemble approaches, CP does not assume any specific data distribution, only that the data are **exchangeable** (i.i.d. is a common special case).

The central guarantee is that for a user-chosen error rate $\alpha \in (0, 1)$,

$$\mathbb{P}\Big(Y_{\text{true}} \notin \hat{C}(X)\Big) \leq \alpha,$$

where $\hat{C}(X)$ is the prediction set for input X. In words: with probability at least $1 - \alpha$, the prediction set will contain the true label or value.

2 Core Principle and Intuition

The idea is to measure how "typical" or "atypical" a new example looks compared to past data. Instead of producing a single prediction (which may be over- or under-confident), CP produces a set of possible outcomes that is guaranteed to contain the truth at least $1 - \alpha$ of the time.

Key steps:

1. Split data into a training set (for fitting the base model) and a calibration set (for calibration).
2. Train any base model f (regressor, classifier, neural network, etc.).
3. Define a **nonconformity score** $s(x, y)$, which measures how unusual (x, y) is relative to the model.
4. Compute nonconformity scores for all calibration points.
5. Take a high quantile of these scores as the threshold.
6. For a new input, include in the prediction set all labels/values whose score is below the threshold.

3 Nonconformity vs. Similarity Scores

A central concept is the **nonconformity score**, which quantifies how "strange" or "non-conforming" a point looks.

- Regression: $s(x, y) = |y - f(x)|$ (absolute residual).
- Classification: $s(x, y) = 1 - \hat{p}(y|x)$ (one minus predicted probability of the true class).

Why not use similarity? If we used a similarity score (high = typical, low = strange), we would need to invert all comparisons to identify outliers. Using nonconformity is natural: larger scores automatically correspond to stranger examples. This makes it straightforward to control the "tail risk" by picking a quantile threshold.

Analogy: Grading exam answers:

- Similarity view: "This answer is 90% similar to the key."
- Nonconformity view: "This answer makes 10% mistakes."

To set a rule like "reject the worst 5% of answers," it is more natural to work with mistakes (nonconformity).

4 Role of α

The parameter α is the **user-chosen error rate**:

$$\mathbb{P}\Big(Y_{\text{true}} \in \hat{C}(X)\Big) \geq 1 - \alpha.$$

- Smaller α (e.g. 0.05) means you want to be wrong only 5% of the time. Prediction sets are larger/wider.
- Larger α (e.g. 0.2) means you accept being wrong 20% of the time. Prediction sets are smaller/tighter.

Thus, α directly controls the **tradeoff between coverage and efficiency**. It is not the same as a Bayesian posterior probability; instead it is a distribution-free guarantee.

5 Quantile Selection in Practice

To achieve the target error rate α, we use the nonconformity scores from the calibration set:

1. Suppose calibration set size is n.
2. Sort the scores: $s_{(1)} \leq s_{(2)} \leq \cdots \leq s_{(n)}$.
3. Choose threshold $q = s_{(k)}$ where

$$k = \lceil (1 - \alpha)(n + 1) \rceil .$$

This ensures finite-sample validity.

Regression Example

Calibration residuals:

$$\{0.5, 1.2, 0.9, 2.1, 1.8, 0.3, 1.1, 2.5, 0.7, 1.4\}.$$

Sorted:

$$\{0.3, 0.5, 0.7, 0.9, 1.1, 1.2, 1.4, 1.8, 2.1, 2.5\}.$$

With $\alpha = 0.1$ and $n = 10$:

$$k = \lceil 0.9 \times 11 \rceil = \lceil 9.9 \rceil = 10.$$

Threshold $q = 2.5$. If $\hat{y} = 8.0$, prediction interval is $[8.0 - 2.5,\ 8.0 + 2.5] = [5.5,\ 10.5]$.

Classification Example

Suppose calibration predicted probabilities for the true class are

$$\{0.90, 0.70, 0.60, 0.50, 0.40, 0.30\}.$$

Scores (1 minus):

$$\{0.10, 0.30, 0.40, 0.50, 0.60, 0.70\}.$$

Sorted: same order.

With $\alpha = 0.1$, $n = 6$:

$$k = \lceil 0.9 \times 7 \rceil = \lceil 6.3 \rceil = 7,$$

so we take the largest score 0.70.

For a new input with predicted probabilities

$$p(A) = 0.65,\ p(B) = 0.25,\ p(C) = 0.10,$$

scores are $0.35, 0.75, 0.90$. Keep labels with score ≤ 0.70. Prediction set $= \{A\}$.

If instead $p(A) = 0.40,\ p(B) = 0.35,\ p(C) = 0.25$, scores are $0.60, 0.65, 0.75$. Prediction set $= \{A, B\}$.

6 Key Takeaways

- Conformal prediction produces set-valued predictions with distribution-free coverage guarantees.
- Nonconformity scores measure "strangeness" and make it easy to threshold extreme cases.
- The parameter α is the miscoverage tolerance; $1 - \alpha$ is the coverage guarantee.
- Prediction sets adapt to model quality: narrow when the model is accurate, wide when the model is uncertain.

7 Conformal Prediction for Time Series

While conformal prediction provides distribution-free coverage guarantees under exchangeability, time series data violate this assumption due to temporal dependence and potential non-stationarity. Extending CP to time series requires careful handling of these dependencies.

Challenges

- **Dependence:** Observations at different time points are correlated, so the usual permutation-based arguments for validity no longer hold directly.
- **Non-stationarity:** Time series may drift or exhibit changing variance, requiring adaptive calibration.
- **Prediction horizon:** For h-step ahead forecasts, uncertainty grows with horizon length.

Approaches

Several strategies have been proposed to adapt CP to time series:

- **Sliding-window conformal prediction:** Use only the most recent w observations as the calibration set. This enforces a form of "local exchangeability" and adapts to drift.
- **Block conformal prediction:** Partition the series into blocks and treat each block as approximately exchangeable, capturing short-range dependencies.
- **Adaptive/weighted conformal prediction:** Weight calibration residuals by recency or importance, giving more influence to recent dynamics.
- **Model-based CP:** Combine CP with probabilistic forecasters (e.g., autoregressive models, RNNs, Transformers) to generate residual-based nonconformity scores that account for temporal structure.

Example: Sliding-window CP

Suppose we want to forecast Y_{t+1} from features X_t. We train a predictive model f on past data, then compute nonconformity scores on a moving calibration window:

$$s_i = |Y_i - f(X_{i-1})|, \quad i = t - w + 1, \ldots, t.$$

We then compute the $(1-\alpha)$-quantile of $\{s_i\}$ and form the prediction interval

$$\hat{C}_{t+1} = \big[f(X_t) - q_{1-\alpha},\, f(X_t) + q_{1-\alpha}\big].$$

As t advances, the calibration window updates, adapting to non-stationary trends.

Applications

Recent work has successfully applied conformal prediction to time series forecasting in domains such as:

- **Traffic forecasting:** Providing uncertainty bands around flow predictions on road networks.
- **Energy demand forecasting:** Quantifying risk in power grid load prediction.
- **Finance:** Forecasting with distribution-free confidence intervals under non-stationarity.

Related Work

- C. Xu and Y. Xie. "Conformal Prediction Interval for Dynamic Time-Series." In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- I. Gibbs and E. Candès. "Adaptive Conformal Inference Under Distribution Shift." In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- A. N. Angelopoulos and S. Bates. "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification." arXiv:2107.07511, 2021.

These works extend CP to dependent data by developing rigorous guarantees under relaxed assumptions (e.g., mixing conditions, block exchangeability) and by designing adaptive calibration strategies.